

A Statistic Based Time Skew Calibration Method for Time-Interleaved ADCs

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Abstract— In this paper, a statistic based time skew calibration method for time-interleaved ADCs is presented. By comparing the mean value of the multiplication of signals in two adjacent channels, the time skew can be estimated. Subsequently, a capacitor array based digitally controlled delay block placed in sampling clock path is adopted to compensate the time skew. In addition, the precision of calibration is further improved through using a monotonic small capacitor array. In a 4-channel 1GS/s 12-bit TI-ADC system, the spurious free dynamic range (SFDR) can be improved to 77.5dB with 0.25ps LSB in the digitally controlled delay block.

I. INTRODUCTION

IN many mix-signal systems, analog-to-digital converters (ADCs) play an important role for digitizing analog signal. To design high-speed and high-resolution ADCs, multi-channel ADC structure is adopted in numerous applications [1-3]. And time-interleaved ADCs (TI-ADCs) are the most frequently used multi-channel ADC structure, as the sampling rate of TI-ADCs can be boosted multiple times. However, due to process variation, there are mismatches between the channels, such as offset mismatch, gain mismatch, time skew and bandwidth mismatch, which result in the performance degradation. Many approaches have been published to mitigate the mismatches in timed-interleaved ADCs [3-4][9-13]. In [4], a multi-channel filter was utilized to compensate time skew and bandwidth mismatch. However, no mismatch parameter estimation block is used and the filter coefficients are fixed. It therefore cannot be adjusted with environmental changing. It was presented in [5] that the reported background calibration techniques can be categorized mainly into three types: the correlation-based approach [3], the equalization-based approach and statistics-based ones.

To compensate the time skew existed in TI-ADCs effectively, a statistic timing-skew calibration technique is presented in this paper, which includes estimating and calibrating the time skew. The remainder of the paper is divided into four sections. Section II gives a brief introduction of mismatches in TI-ADCs. In Section III, the proposed calibration method as well as a monotonic capacitor based digitally controlled delay block are described. Section

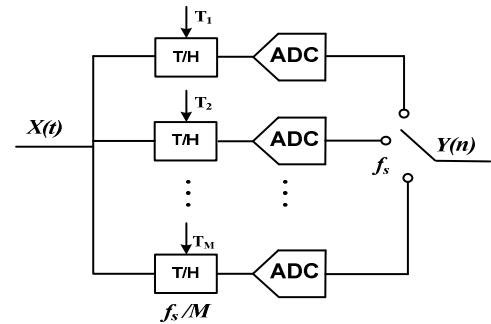


Fig. 1. The structure of time-interleaved ADCs.

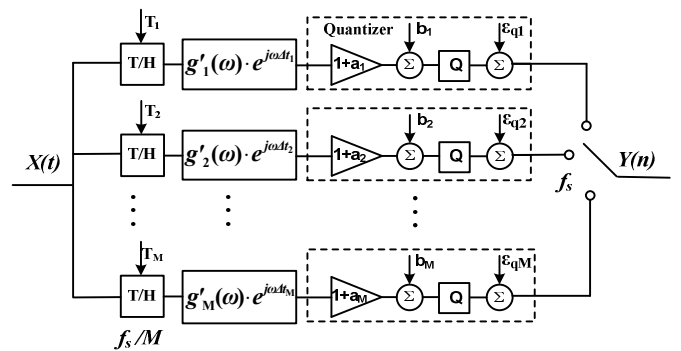


Fig. 2. The model of time-interleaved ADCs, including bandwidth mismatch, offset mismatch, gain mismatch and time skew.

IV shows the design example and simulation results, and finally conclusion is drawn in Section V.

II. MISMATCHES IN TIME-INTERLEAVED ADCS

Fig. 1 presents the block diagram of M-channel TI-ADCs. The input signal is sampled by multi-phase clock with period MT_s and combined in digital domain with a digital clock with period T_s . The model of M-channel TI-ADCs is shown in Fig. 2, including quantization noise ϵ_{qm} , offset mismatch b_m , gain mismatch a_m , bandwidth mismatch $g'_m(\omega)$ and time skew Δt_m . Detailed discussion of mismatches can be found in [6-7]. For the normal priority of calibration, offset mismatch is

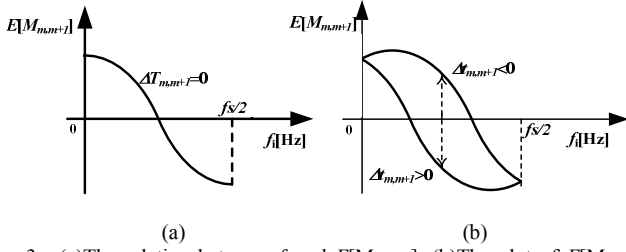


Fig. 3. (a) The relation between f_i and $E[M_{m,m+1}]$. (b) The plot of $E[M_{m,m+1}]$ with $\Delta t_{m,m+1} > 0$ and $\Delta t_{m,m+1} < 0$.

compensated first, followed sequentially by the gain mismatch, bandwidth mismatch calibration, and finally the time skew, which can be calibrated with different techniques [9-13]. Offset mismatch and gain mismatch can be calibrated easily [1][7]. A digital compensation filter based bandwidth mismatch calibration approach is proposed in [14]. However, due to correlation between channels, time skew cannot be calibrated easily. Though it can be compensated by post digital processing [3], the performance cannot satisfy the SFDR requirement at high input frequency. In [8], a digital filter bank is adopted to compensate all the mismatches in TI-ADCs, but it does not propose channel mismatches estimation techniques, obtaining the mismatches through instruments. There is also a background time skew calibration method published in [9], which, however, is only suitable for input signals with wide bandwidth and extra slicers are needed for the sample time error estimation. In [11], an average value calculation of all the channels is employed. Unfortunately, it is not feasible for large number of channels implementation.

III. STATISTIC CALIBRATION TECHNIQUE FOR TI-ADCs

A. Statistic Based Calibration Technique

In this section, the proposed statistic calibration method is introduced. Assumes that the input signal is

$$X(t) = \sum_{i=0}^{K-1} A_i \sin(2\pi f_i t + \theta_i), \quad (1)$$

which consists of K frequency components. Therefore, the multiplication of two adjacent sub-ADC outputs $x'_m(n)$ and $x'_{m+1}(n)$ is

$$\begin{aligned} M_{m,m+1}(n) &= x'_m(n) \cdot x'_{m+1}(n) \\ &= \sum_{i=0}^{K-1} A_i \sin(2\pi f_i M T_s n + \theta_i) \cdot \sum_{i=0}^{K-1} A_i \sin(2\pi f_i ((Mn+1)T_s + \Delta t_{m,m+1}) + \theta_i) \\ &= \frac{1}{2} \sum_{i=0}^{K-1} A_i^2 \cos(2\pi f_i (T_s + \Delta t_{m,m+1})) \\ &\quad - \frac{1}{2} \sum_{i=0}^{K-1} A_i^2 \cos(2\pi f_i (2nMT_s + T_s + \Delta t_{m,m+1}) + 2\theta_i) \\ &\quad + \sum_{\substack{i1,i2=0 \\ i1 \neq i2}}^{K-1} A_{i1} A_{i2} \sin(2\pi f_{i1} M T_s n + \theta_{i1}) \cdot \sin(2\pi f_{i2} (M T_s n + T_s + \theta_{i2})), \end{aligned}$$

(2)

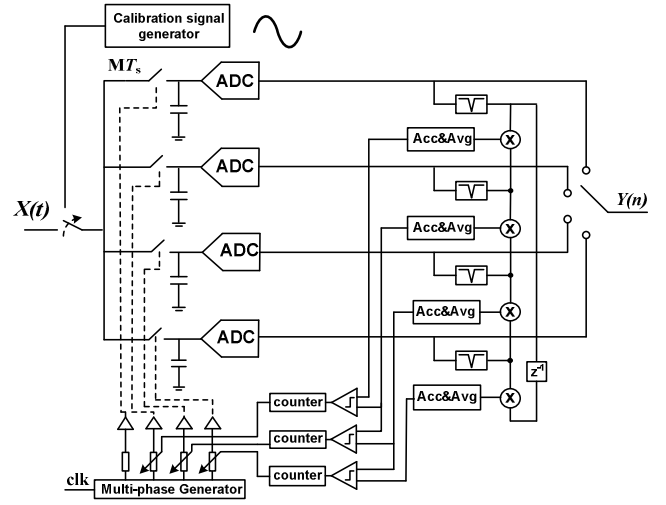


Fig. 4. The proposed calibration structure for 4-channel TI-ADCs.

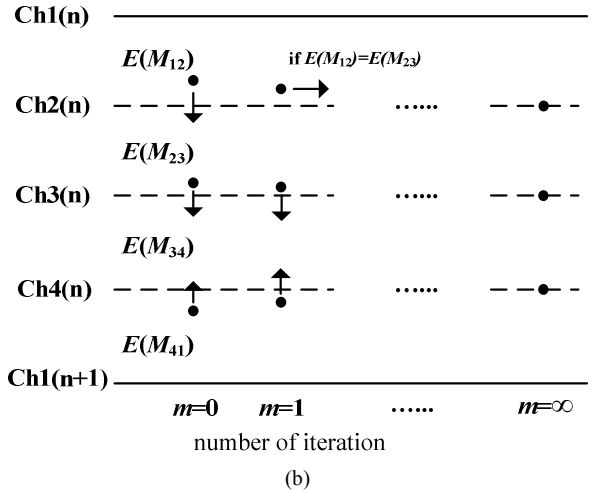
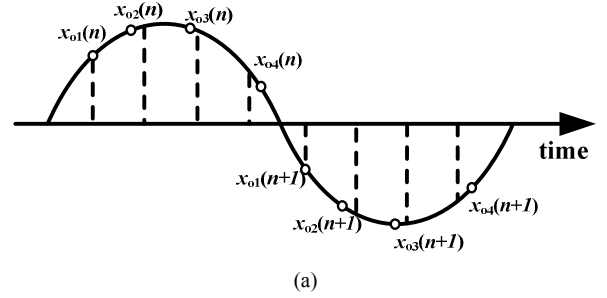


Fig. 5. Illustration of calibration process. (a) Input signal with time skew. (b) The process of convergence.

where $\Delta t_{m,m+1}$ is the time skew between channel m and channel $m+1$. If $f_{i1} \pm f_{i2} \neq k/MT_s$ ($k=0, 1, 2, \dots, M/2$) and $f_i \neq k/2MT_s$ ($k=0, 1, 2, \dots, M/2$), the mean value of $M_{m,m+1}(n)$ is

$$E[M_{m,m+1}] = \frac{1}{2} \sum_{i=0}^l A_i^2 \cos(2\pi f_i (T_s + \Delta t_{m,m+1})). \quad (3)$$

It can be seen from (3) that $E[M_{m,m+1}]$ is related with the time skew. In the first Nyquist zone of which the range of f_i is $[0, f_s/2]$, the relation of f_i and ideal mean value $E[M_{m,m+1}]$ is

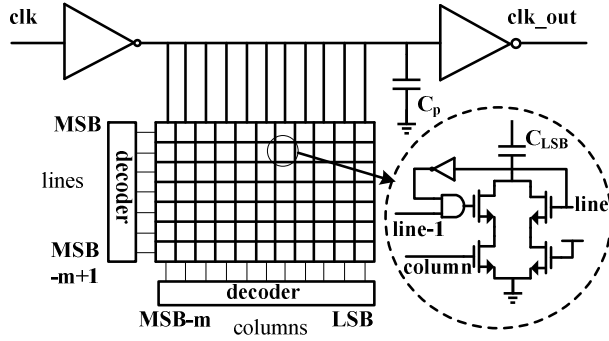


Fig. 6. The capacitor array based digitally controlled delay block.

drawn in Fig. 3(a). In Fig. 3(a), it is seen that the relation of f_i and $E[M_{m,m+1}]$ is monotonic, a necessary condition for time skew error detection. In other words, the value of $E[M_{m,m+1}]$ is able to indicate the polarity of time skew. In Fig. 3(b), two conditions ($\Delta t_{m,m+1} < 0$ and $\Delta t_{m,m+1} > 0$) are drawn when $\Delta t_{m,m+1}$ is non-zero value. Normally, the mean value is estimated by averaging N samples. Hence, a large N can increase the accuracy of calibration. Additionally, to maintain an effective average calculation, a notch filter is employed to filter out the frequency components $k/2MT_s$ ($k=0, 1, 2, \dots, M$).

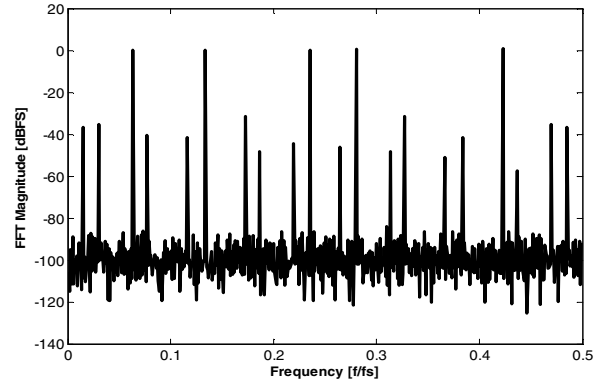
B. Calibration Operation and Digitally Controlled Delay

The block diagram of a 4-channel TI-ADC including the proposed calibration technique is shown in Fig. 4. The calibration structure is a close-loop iterative system. During the calibration, the input is connected a calibration signal generator for which there is no linearity requirement. The samples from sub-ADC of each channel goes through a notch filter first, and multiplication is operated between every two adjacent channels, then output value of the accumulator and average block is the approximation of mean value $E[M_{m,m+1}]$ mentioned above. Next, every two adjacent $E[M_{m,m+1}]$ are compared to generate one bit output, which makes the counter add or subtract one LSB (T_{step}), which is the adjustment resolution of the digital controlled delay block. And the value in the counter controls the phase of sampling clock. There is a non-ideal sampling case of a 4-channel TI-ADC in Fig. 5(a), where channel 1 is the reference channel. It can be seen from Fig. 5(a) that $\Delta t_{12} < 0$, $\Delta t_{23} < 0$ ($|\Delta t_{12}| > |\Delta t_{23}|$), $\Delta t_{34} > 0$ and $\Delta t_{41} < 0$. The operation process is illustrated in Fig. 5(b), where m is the number of iteration and arrows show the direction of phase will change. When $m=0$, $E(M_{12}) > E(M_{23})$, so the phase of channel 2 increase by one LSB (T_{step}). Same operations are conducted in channel 3 and channel 4. And if $E(M_{12})=E(M_{23})$, the delay of channel will be stay until next iteration. In addition, the process of calibration should be demonstrated to be convergent. First, defining the $EM_{max}(m)$ and $EM_{min}(m)$ as

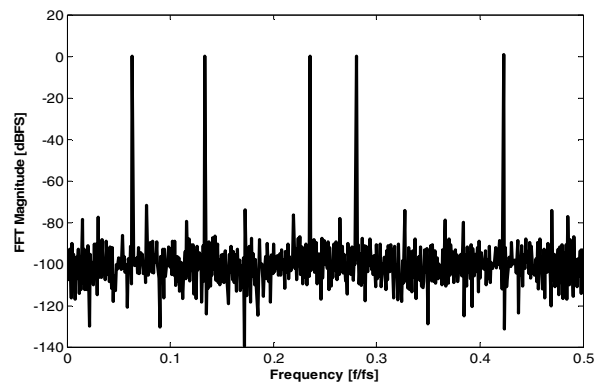
$$EM_{max}(m) = \max_m |E(M_{12}), E(M_{23}), E(M_{34}), E(M_{41})| \quad (4)$$

and

$$EM_{min}(m) = \min_m |E(M_{12}), E(M_{23}), E(M_{34}), E(M_{41})|, \quad (5)$$



(a)



(b)

Fig. 7. The 2048-point FFT of output signal. (a) Before calibration. (b) After calibration.

respectively. According to algorithm illustrated in Fig. 5(b), the value of $EM_{max}(m)$ will always be decreased (or unchanged), and $EM_{min}(m)$ will always be increased (or unchanged). In another way, it means $EM_{max}(m+1) \leq EM_{max}(m)$ and $EM_{min}(m+1) \geq EM_{min}(m)$. Therefore, the conclusion can be drawn as

$$\lim_{m \rightarrow \infty} EM_{max}(m) = \lim_{m \rightarrow \infty} EM_{min}(m), \quad (6)$$

which means as m trends to infinity, all the phase delay between two adjacent channels become the same. In other words, there is no time skew. The time needed for convergence T_{con} depends on the variation of time skew Δt_M and number of samples N for the estimation of mean value. It can be represented as

$$T_{con} = \frac{\max |\Delta t_M|}{T_{step}} \cdot N \cdot T_s, \quad (7)$$

where T_s is the sampling period of the TI-ADC system.

As shown in Fig. 6, there is a capacitor array based digitally controlled delay block. The number of C_{LSB} connected to output of clock buffer is controlled by a binary code, of which the MSBs bits control the lines and LSBs control the columns. With this arrangement, the phase adjustment will be monotonic, which is more important than linearity of the capacitor array. C_p is the parasitic capacitor.

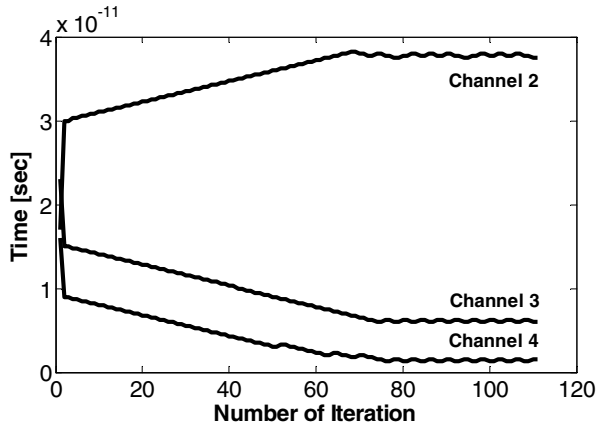


Fig. 8. Convergence of the optimized time delay values.

IV. SIMULATION RESULTS

In this section, a behavioral level 4-channel 1GS/s 12-bit TI-ADC system is designed to demonstrate the proposed calibration technique. The system is similar with the block diagram in Fig. 4. The transfer function of notch filter is

$$H(z) = \frac{1 - z^{-2}}{1 - p^2 z^{-2}}, \quad (8)$$

where $p=0.99$. The input signal $x(t) = \sin(2\pi f_{i1}t + \theta_1) + \sin(2\pi f_{i2}t + \theta_2) + \sin(2\pi f_{i3}t + \theta_3) + \sin(2\pi f_{i4}t + \theta_4) + \sin(2\pi f_{i5}t + \theta_5)$, where $\theta_k=0$ ($k=1,2,3,4,5$), $f_{i1}=63.48\text{MHz}$, $f_{i2}=133.8\text{MHz}$, $f_{i3}=204.1\text{MHz}$, $f_{i4}=280.27\text{MHz}$, and $f_{i5}=422.85\text{MHz}$. Setting the initial time skew $[\Delta t_1, \Delta t_2, \Delta t_3, \Delta t_4] = [20\text{ps}, 2.1\text{ps}, 33.9\text{ps}, 38.4\text{ps}]$ and LSB of the digitally controlled delay array is 0.25ps. The number of samples N needed to accumulate and average is set to 10000. The 2048-point fast Fourier transform (FFT) of output signal before calibration is shown in Fig. 7(a), while Fig. 7(b) shows FFT of output signal after calibration. It can be seen from Fig. 7 that the SFDR is improved to 77.5dB after calibration with 0.25ps LSB (T_{step}). By fixing the delay of channel 1 as 20ps, the delay of channel m ($m=2, 3, 4$) can be adjusted accordingly. The convergence process of calibrated value in digitally controlled delay array is presented in Fig. 8, where the calibration is finished after about 75 times of iteration. Finally, the optimized time delay value is [20ps, 37.5ps, 6ps, 1.5ps], which equalize the delay of each channel around 40ps.

V. CONCLUSION

In this paper, a statistic based calibration technique for M-channel TI-ADC is proposed. Compared with the previous published techniques, it provides an effective time skew detection method, and with the digitally controlled delay block, the precision of the calibration can be further improved by decreasing the size of unit capacitor in the delay block. In addition, the calibration technique can be readily extended to TI-ADCs with more than four-channels.

REFERENCES

- [1] C. C. Hsu, F. C. Huang, C. Y. Shih, C. C. Huang, Y. H. Lin, C. C. Lee, and B. Razavi, "An 11b 800MS/s Time-Interleaved ADC with Digital Background Calibration," in *IEEE ISSCC Dig. Tech. Papers*, 2007, pp. 464-465.
- [2] S. K. Gupta, M. A. Inerfield, and J. Wang, "A 1-GS/s 11-bit ADC With 55-dB SNDR, 250-mW Power Realized by a High Bandwidth Scalable Time-Interleaved Architecture," *IEEE J. Solid-State Circuits*, vol. 41, no.12, pp. 2650-2657, Dec. 2006.
- [3] L. Chi Ho, P. J. Hurst and S. H. Lewis, "A Four-Channel Time-Interleaved ADC with Digital Calibration of Inter-channel Timing and Memory Errors," *IEEE J. Solid-State Circuits*, vol. 45, no. 10, pp. 2091-2103, 2010.
- [4] Y. C. Lim, Y. X. Zou, J. W. Lee, S. C. Chan, "Time-Interleaved Analog-to-Digital-Converter Compensation Using Multichannel Filters," *IEEE Trans. Circuit and Syst. I: Regular papers*, vol.56, no.10, pp.2234-2247, Oct. 2009.
- [5] C. Tsang, Y. Chiu, J. Vanderhaegen, S. Hoyos, C. Chen, R. Brodersen and B. Nikolic, "Background ADC calibration in digital domain," in *Proc. IEEE CICC*, 2008, pp.301-304.
- [6] N. Kurosawa, H. Kobayashi, K. Maruyama, H. Sugawara and K. Kobayashi, "Explicit analysis of channel mismatch effects in time-interleaved ADC systems," *IEEE Trans. Circuits Syst. I: Fundam. Theory Appl.*, vol. 48, no. 3, pp. 261-271, Mar. 2001.
- [7] Tsung-Heng Tsai; Hurst, P.J.; Lewis, S.H.;, "Correction of Mismatches in a Time-Interleaved Analog-to-Digital Converter in an Adaptively Equalized Digital Communication Receiver," *IEEE Trans. Circuits Syst. I: Regular Papers*, vol.56, no.2, pp.307-319, Feb. 2009.
- [8] S. Munkyo, M. J. W. Rodwell, and U. Madhow, "Comprehensive digital correction of mismatch errors for a 400-msamples/s 80-dB SFDR time-interleaved analog-to-digital converter," *IEEE Trans. on Microwave Theory and Techniques*, vol. 53, pp. 1072-1082, 2005.
- [9] A. Haftbaradaran, K.W. Martin, "A Background Sample-Time Error Calibration Technique Using Random Data for Wide-Band High-Resolution Time-Interleaved ADCs," *IEEE Trans. Circuits Syst. II: Express Briefs*, vol.55, no.3, pp.234-238, March 2008.
- [10] Camarero, D, Ben Kalaia, K, J. F. Naviner, P. Loumeau, "Mixed-Signal Clock-Skew Calibration Technique for Time-Interleaved ADCs," *IEEE Trans. Circuits Syst. I: Regular Papers*, vol.55, no.11, pp.3676-3687, Dec. 2008.
- [11] Chung-Yi Wang; Jieh-Tsong Wu; , "A Multiphase Timing-Skew Calibration Technique Using Zero-Crossing Detection," *IEEE Trans. Circuits Syst. I: Regular Papers*, vol.56, no.6, pp.1102-1114, June 2009.
- [12] Chung-Yi Wang; Jieh-Tsong Wu; , "A background timing-skew calibration technique for time-interleaved analog-to-digital converters," *IEEE Trans. Circuits Syst. II: Express Briefs*, vol.53, no.4, pp. 299-303, April 2006.
- [13] H. Jin and E. K. F. Lee, "A digital-background calibration technique for minimizing timing-error effects in time-interleaved ADCs," *IEEE Trans. Circuits Syst. II: Analog Digit. Signal Process.*, vol. 47, no. 7, pp. 603-613, Jul. 2000.
- [14] T. H. Tsai, P. J. Hurst, and S. H. Lewis, " Bandwidth Mismatch and Its Correction in Time-Interleaved Analog-to-Digital Converters," *IEEE Trans. Circuits Syst. II: Express Briefs*, vol.53, no.10, pp.1133-1137, Oct. 2006.